(1)

$f(x)$ fonksiyonunun ve birinci, ikinci ve üçüncü mertebeden türevlerinin bazı x noktalarında aldığı değerler aşağıdaki tabloda verilmiştir. $f(x)$ fonksiyonunun $x = 1$ civarında 3. mertebeden Taylor polinomu aşağıdakilerden hangisidir?

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-2	1	4
1	2	-3	3	-2
2	-1	1	4	5

→ $x=1$ için bu satır önemli :)

$$P_3(x) = f(1) + f'(1)\cdot(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3$$

$$= 2 - 3(x-1) + \frac{3}{2}(x-1)^2 - \frac{2}{6}(x-1)^3$$

Cevap C

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(2) $\sum_{n=0}^{\infty} \frac{(x+1)^{2n}}{9^n}$ serisinin toplamı ve yakınsaklık aralığı aşağıdakilerden hangisinde verilmiştir?

$$\sum_{n=0}^{\infty} \frac{(x+1)^{2n}}{9^n} = \underbrace{1}_{a} + \frac{(x+1)^2}{9} + \frac{(x+1)^4}{9^2} + \frac{(x+1)^6}{9^3} + \dots$$

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$r = \frac{(x+1)^2}{9}$ $\Rightarrow |r|<1$ için yakınsak

$$\left|\left(\frac{x+1}{3}\right)^2\right| < 1 \Rightarrow \frac{|x+1|}{3} < 1 \rightarrow -3 < x+1 < 3$$

$$-4 < x < 2$$

$$\text{Toplam} = \frac{a}{1-r} = \frac{1}{1-\frac{(x+1)^2}{9}} = \frac{9}{9-(x+1)^2}$$

Cevap D

(3) $f(x)=\dfrac{x^4}{x^2+3}=\dfrac{x^4}{3\left(1+\frac{x^2}{3}\right)}$

$\sum_{n=0}^{\infty} x^n=\dfrac{1}{1-x}$ $\xrightarrow{x\to -\frac{x^2}{3}}$ $\sum_{n=0}^{\infty}\dfrac{(-1)^n\cdot x^{2n}}{3^n}=\dfrac{1}{1+\frac{x^2}{3}}$ $\xrightarrow{\frac{x^4}{3}\text{ carp}}$ $\sum_{n=0}^{\infty}\dfrac{(-1)^n\cdot x^{2n+4}}{3^{n+1}}=\dfrac{x^4}{3+x^2}$

Cevap A

(4) $\sum_{n=0}^{\infty}\dfrac{(-1)^n\pi^{2n+1}}{3(2n+1)!\,9^n}$ serisinin toplamı kaçtır?

$9^n=3^{2n}$

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Genel terimdeki faktöriyel, n size Sinx, Cosx açılımlarını hatırlatsın ☺

$\sum_{n=0}^{\infty}\dfrac{(-1)^n\cdot\pi^{2n+1}}{(2n+1)!\,3^{2n+1}}=\sum_{n=0}^{\infty}\dfrac{(-1)^n}{(2n+1)!}\left(\dfrac{\pi}{3}\right)^{2n+1}$

$\sin x=\sum_{n=0}^{\infty}(-1)^n\cdot\dfrac{x^{2n+1}}{(2n+1)!}$ $\xrightarrow{x=\frac{\pi}{3}}$ $\sin\frac{\pi}{3}=\sum_{n=0}^{\infty}\dfrac{(-1)^n}{(2n+1)!}\left(\dfrac{\pi}{3}\right)^{2n+1}$

$\frac{\sqrt{3}}{2}$

Cevap B

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(5) $F(x)=\int_0^x\dfrac{1-e^{-t^2}}{t^2}\,dt$ fonksiyonunun Maclaurin serisi?

$e^x=1+x+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}+\cdots$

$e^{-t^2}=1-t^2+\dfrac{t^4}{2!}-\dfrac{t^6}{3!}\cdots$ $\rightarrow$ $\dfrac{1-e^{-t^2}}{t^2}=\dfrac{\cancel{1}-\left[\cancel{1}-t^2+\frac{t^4}{2!}-\frac{t^6}{3!}\cdots\right]}{t^2}=1-\dfrac{t^2}{2!}+\dfrac{t^4}{3!}\cdots$

$F(x)=\int_0^x\left(1-\dfrac{t^2}{2!}+\dfrac{t^4}{3!}\cdots\right)dt=t-\dfrac{t^3}{3.2!}+\dfrac{t^5}{5.3!}\cdots\Big|_0^x=x-\dfrac{x^3}{3.2!}+\dfrac{x^5}{5.3!}-\cdots$

$n=0$, $n=1$, $n=2$

$=\sum_{n=0}^{\infty}(-1)^n\cdot\dfrac{x^{2n+1}}{(2n+1)(n+1)!}$

Cevap B

(6) 3. mertebe Taylor Polinomu kullanılarak yapılacak yaklaşık değer hesaplamasında $\ln(1,2)$ sayısının yaklaşık değeri aşağıdakilerden hangisidir?

a) 0,182 b) 0,222 c) 0,184 d) 0,22

$\ln(1,2) \approx ?$ $f(x)=\ln x$ $a=1$ olsun.

$$P_3(x) = f(1) + f'(1)(x-1) + f''(1)\frac{(x-1)^2}{2!} + \frac{f'''(1)}{3!}(x-1)^3$$

$f(x)=\ln x \to f'(x)=\frac{1}{x} \to f''(x)=-\frac{1}{x^2} \to f'''(x)=\frac{2}{x^3}$

$f(1)=0$ $f'(1)=1$ $f''(1)=-1$ $f'''(1)=2$

$$P_3(x) = (x-1) - \frac{(x-1)^2}{2} + \frac{2}{3!}(x-1)^3$$

$$f(1,2) \approx P_3(1,2) = (1,2-1) - \frac{(1,2-1)^2}{2} + \frac{1}{3}(1,2-1)^3$$

$$= 0,2 - 0,02 + \frac{0,008}{3} = 0,182$$

Cevap A

(7) $I(x) = \int_0^x \frac{1-\cos t}{t^2}\,dt$ fonksiyonunun Maclaurin serisi?

a) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^{2n+1}}{(2n+1)\cdot(2n+2)!}$ b) $\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n+1}}{(2n+1)\cdot(2n+2)!}$ c) $\sum_{n=0}^{\infty} \frac{(-1)^n \cdot x^{2n}}{2n\cdot(2n)!}$

$$\cos t = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots \Rightarrow \frac{1-\cos t}{t^2} = \frac{t - \left[1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots\right]}{t^2}$$

$$= \frac{1}{2!} - \frac{t^2}{4!} + \frac{t^4}{6!} \cdots$$

$$I(x) = \int_0^x \left(\frac{1}{2!} - \frac{t^2}{4!} + \frac{t^4}{6!} \cdots\right)dt = \frac{t}{2!} - \frac{t^3}{3\cdot 4!} + \frac{t^5}{5\cdot 6!} \cdots \Big|_0^x = \frac{x}{2!} - \frac{x^3}{3\cdot 4!} + \frac{x^5}{5\cdot 6!} \cdots$$

($\to n=0$, $\to n=1$, $\to n=2$)

$$= \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)\cdot(2n+2)!}$$

Cevap B

8

$$\int_0^{\frac{1}{3}} \frac{\ln(1-x)}{x} dx = ?$$

a) $1 + \frac{1}{2 \cdot 2!} + \frac{1}{3 \cdot 3!} + \cdots + \frac{1}{n \cdot n!} + \cdots$

b) $-\frac{1}{3} - \frac{1}{4 \cdot 3^2} - \frac{1}{9 \cdot 3^3} - \cdots - \frac{1}{n^2 \cdot 3^n} - \cdots$

c) $\frac{1}{2} - \frac{3}{4!} + \frac{5}{6!} - \cdots + (-1)^n \frac{2n-1}{(2n)!} + \cdots$

d) $1 - \frac{1}{2} + \frac{1}{4} - \cdots + \frac{1}{(-2)^n} + \cdots$

e) $\frac{1}{2} + \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} + \cdots + \frac{1}{n \cdot 2^n} + \cdots$

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ $\xrightarrow{x \to -x}$ $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$

$\frac{\ln(1-x)}{x} = -1 - \frac{x}{2} - \frac{x^2}{3} - \cdots$

$\downarrow$

$$\int_0^{1/3} \left(-1 - \frac{x}{2} - \frac{x^2}{3} - \cdots\right) dx = -x - \frac{x^2}{2 \cdot 2} - \frac{x^3}{3 \cdot 3} \cdots \Big|_0^{1/3}$$

$$= -\frac{1}{3} - \frac{1}{4 \cdot 3^2} - \frac{1}{9 \cdot 3^3} \cdots$$

Ceva B

(9) $\int_0^2 \frac{\cos x - 1}{x}\,dx = ?$

(a) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n}}{(2n).(2n)!}$

(b) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n+2}}{(2n+2).(2n+2)!}$

(c) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n+1}}{(2n+1).(2n+1)!}$

(d) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n-1}}{(2n-1).(2n-1)!}$

(e) $\sum_{n=1}^{\infty} \frac{2^n}{n.n!}$

$$\cos x = \sum_{n=0}^{\infty}(-1)^n \cdot \frac{x^{2n}}{(2n)!} = 1 + \sum_{n=1}^{\infty}(-1)^n \cdot \frac{x^{2n}}{(2n)!}$$

$$\frac{\cos x - 1}{x} = \frac{1 + \sum_{n=1}^{\infty}(-1)^n \cdot \frac{x^{2n}}{(2n)!} - 1}{x} = \sum_{n=1}^{\infty}(-1)^n \cdot \frac{x^{2n-1}}{(2n)!}$$

$$\int_0^2 \frac{\cos x - 1}{x}\,dx = \int_0^2 \left(\sum_{n=1}^{\infty}(-1)^n \cdot \frac{x^{2n-1}}{(2n)!} \right) dx$$

$$= \sum_{n=1}^{\infty}(-1)^n \cdot \frac{x^{2n}}{2n.(2n)!}\Bigg|_0^2 = \sum_{n=1}^{\infty}(-1)^n \cdot \frac{2^{2n}}{2n.(2n)!}$$

(10) $\int_0^2 \frac{\sin x - x}{x}\,dx = ?$

(a) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n}}{(2n).(2n)!}$

(b) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n+2}}{(2n+2).(2n+2)!}$

(c) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n+1}}{(2n+1).(2n+1)!}$

(d) $\sum_{n=1}^{\infty}(-1)^n \frac{2^{2n-1}}{(2n-1).(2n-1)!}$

(e) $\sum_{n=1}^{\infty} \frac{2^n}{n.n!}$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} = x + \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!}$$

$$\frac{\sin x - x}{x} = \frac{x + \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} - x}{x} = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n}}{(2n+1)!}$$

$$\int_0^2 \frac{\sin x - x}{x}\,dx = \int_0^2 \left(\sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n}}{(2n+1)!} \right) dx$$

$$= \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1).(2n+1)!} \Bigg|_0^2 = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{2^{2n+1}}{(2n+1).(2n+1)!}$$